GEAR-Nav

Action-free spatial trajectory refinement via energy-based aggregation

Learning corrections
from geometric energy



Indoor and outdoor robot deployments

Learning from video without target paths



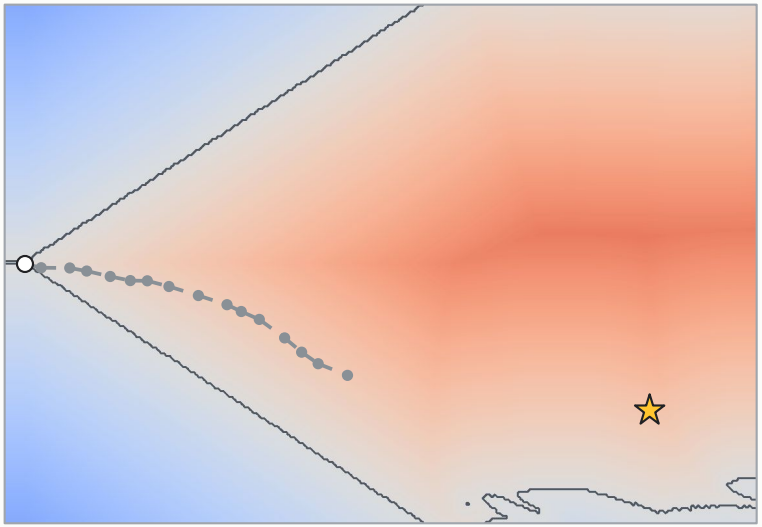
In-the-wild egocentric observation

Geometric supervision for navigation adaptation

Video provides scenes, but no robot actions or target paths.

Learn how a candidate path should change.

Geometry defines the training signal



Candidate path in reconstructed geometry

Goal-directed progress

Reach the destination; discourage retreat.

Footprint clearance

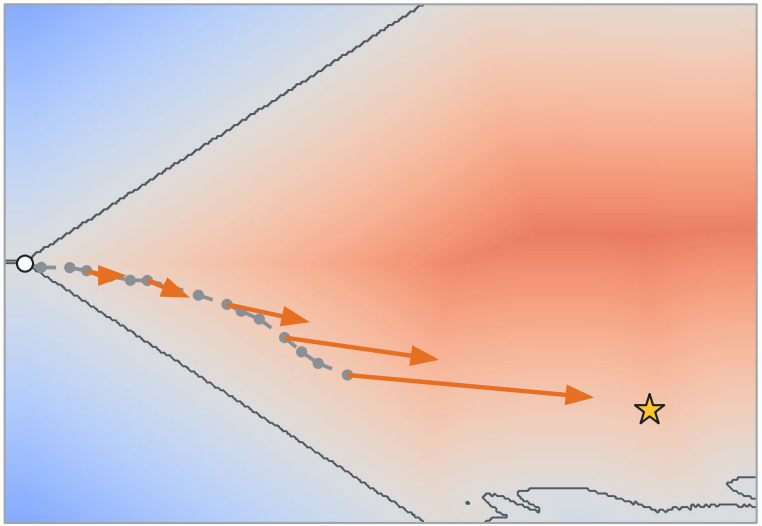
Penalize collisions and close obstacles.

Motion regularity

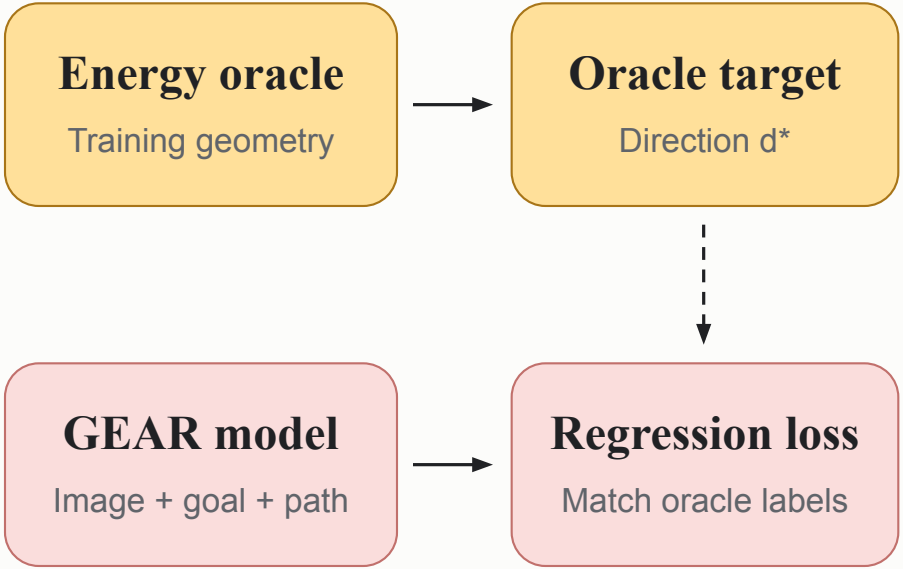
Encourage smooth, goal-facing paths.

$$E(x; g, \Phi) = \sum_q \lambda_q E_q(x; g, \Phi)$$

Learn waypoint corrections

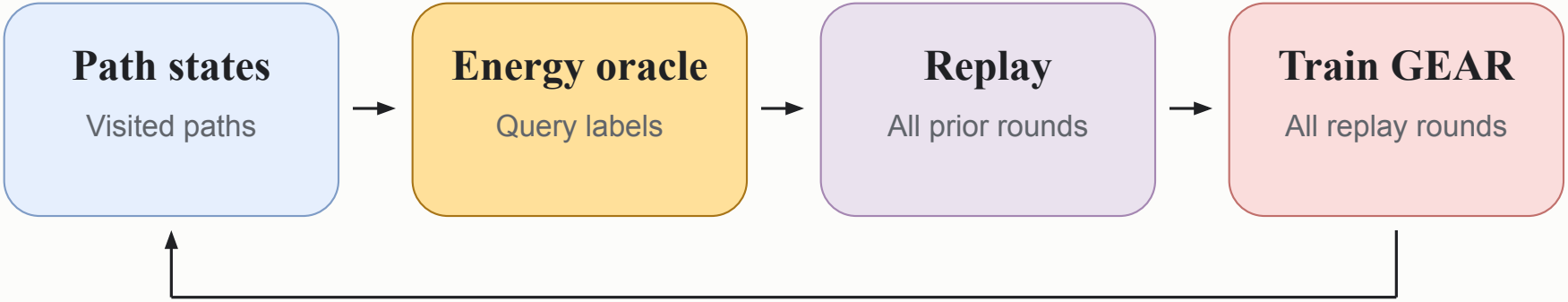


Local energy-descent directions



Supervise the current candidate path.

Aggregate labels along model rollouts



Visit the next candidate paths

B_1 B_2 B_3 … B_r

Retain correction labels across aggregation rounds.

Refine a path from vision



RGB observation

Navigation goal

Pretrained VLA

Visual + goal context

Candidate path

Waypoint expert

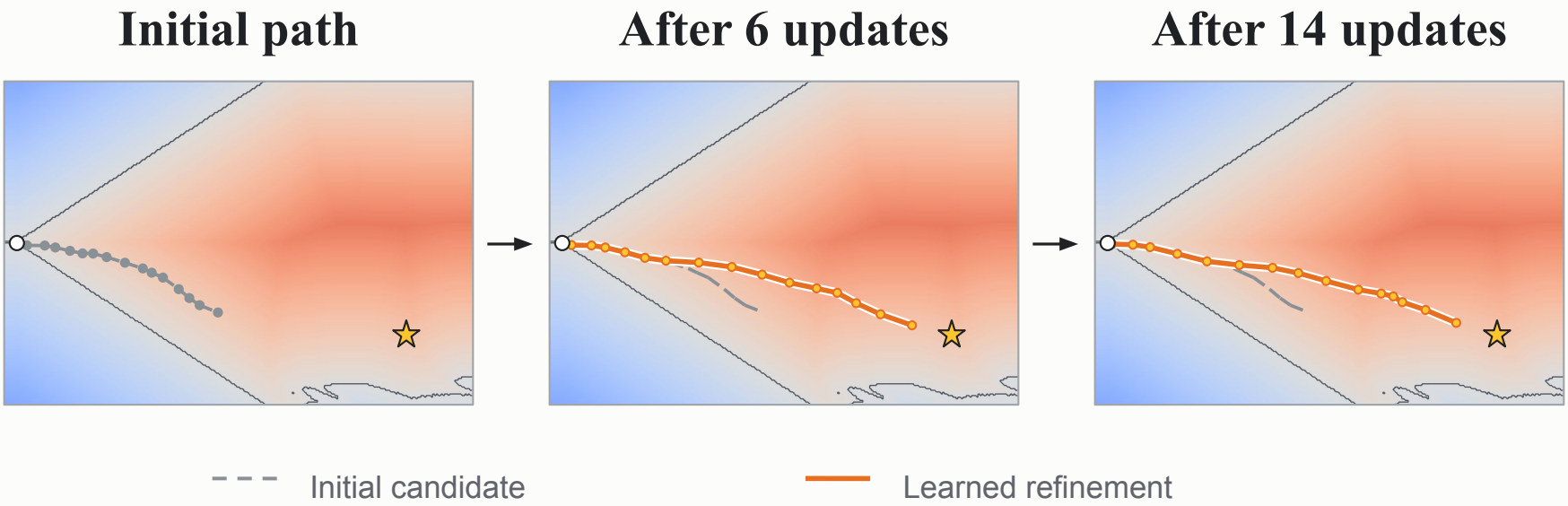
Predict a correction

Bounded path update

Repeat with the updated path

No ESDF or energy queries at deployment

Recorded model-only refinement



ESDF shown for reference; the model does not query it.

REAL-WORLD / SCENARIO 1

Around an intervening chair

GOAL

Person beyond the chair

Representative separate trials

GEAR-Nav



OmniVLA



1× playback

REAL-WORLD / SCENARIO 2

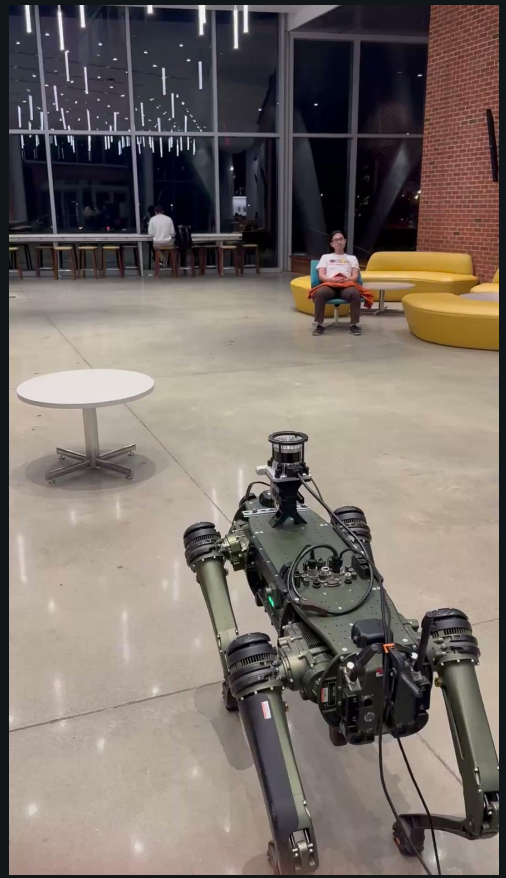
Past a moving person

GOAL

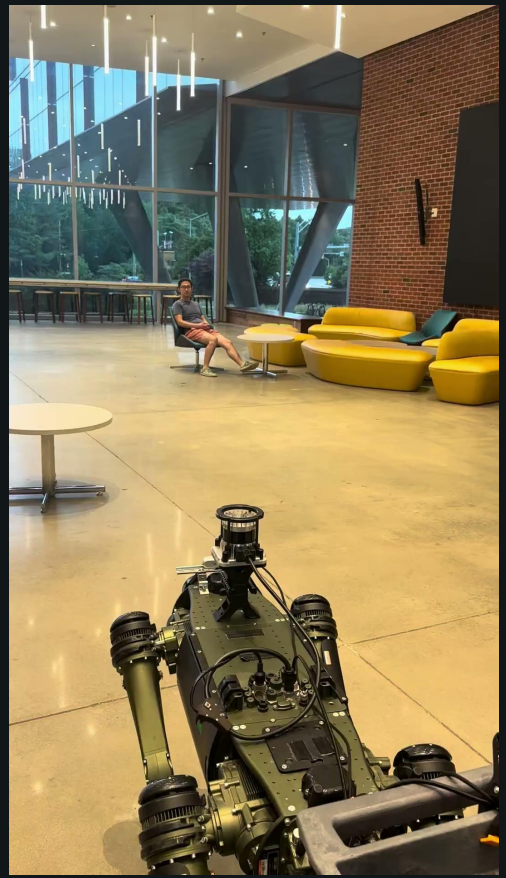
Blue chair in the lounge

Representative separate trials

GEAR-Nav



OmniVLA



1× playback

Scooter goal among pedestrians

SCENARIO 3

GEAR-Nav



OmniVLA



NaVILA



Goal: scooter beyond the pedestrians

Separate trials · 1× playback

Additional robot deployments

Indoor



1× playback

Lounge



1× playback

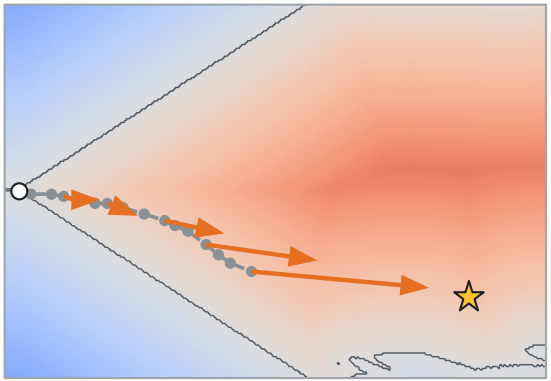
Outdoor



1× playback

From geometric supervision to visual refinement

TRAINING



GEAR-Nav

Correction field

DEPLOYMENT



Energy oracle

Query local corrections

Aggregate labels at model-visited paths

Visual refinement

RGB + goal + candidate path

Training geometry supplies the supervision; the learned field refines paths at deployment.

GEAR-Nav

Action-free spatial trajectory refinement via energy-based aggregation

Geometric supervision during training.
Vision-conditioned refinement at deployment.